Classification of Alzheimer's Disease and Frontotemporal Dementia from Resting-State EEG Using Hand-Crafted Features and Deep Learning

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***Abstract*—Early and accurate differentiation of Alzheimer's disease (AD), frontotemporal dementia (FTD), and healthy controls (HC) from resting-state electroencephalography (EEG) remains an open clinical challenge. We evaluate two complementary pipelines on the publicly available ds004504 dataset [4] (88 subjects, 19 channels, 500 Hz): (1) a hand-crafted feature pipeline with 1 805 features per 20-second epoch (time-domain, frequency-domain, DWT, and inter-channel coherence) combined with SVM, Random Forest, and XGBoost; and (2) a Continuous Wavelet Transform (CWT) scalogram pipeline driving seven deep-learning architectures—including YOLO26-cls variants and a One-vs-Rest (OvR) ensemble—as well as three EEG-native models (EEGNet, ShallowConvNet, SpectrogramCNN). All splits are subject-wise stratified (70/10/20) to prevent data leakage; final scores are reported at segment and subject (majority-vote) level. The best result—88.89% subject-level accuracy and 88.77% macro-F1—is achieved by an OvR YOLO26m-cls ensemble with validation-based decision-threshold optimisation.**

***Index Terms*—Alzheimer's disease, frontotemporal dementia, EEG, CWT scalogram, hand-crafted features, XGBoost, EEGNet, YOLO, one-vs-rest classification**

I. Introduction

Dementia affects over 55 million people worldwide; Alzheimer's disease (AD) accounts for 60–70% of cases, while frontotemporal dementia (FTD) is the second most common young-onset subtype [1]. Clinical differentiation between AD, FTD, and healthy aging is challenging because both diseases share overlapping cognitive profiles in early stages and produce partially overlapping electrophysiological signatures. Electroencephalography (EEG) is a low-cost, non-invasive modality capturing cortical dynamics at high temporal resolution. In AD, EEG is associated with dominant-rhythm slowing, elevated delta/theta power, and reduced alpha/beta activity; FTD shows a distinct pattern of frontal hypoactivation [2], [3].

The ds004504 dataset [4] has rapidly become the standard benchmark for EEG-based dementia classification. Table I summarises all independently verified studies on this dataset. Initial work—DICE-Net [5]—achieved 96.3% binary (AD vs. HC) accuracy and ∼83% 3-class accuracy using a convolution-transformer on STFT spectrograms. Subsequent approaches include: multilayer PLV graph networks (SVM, 89.6% AD/FTD binary) [6]; shallow CNNs with AEC functional connectivity (94.54% binary) [7]; Lyapunov exponents + SVM (90.2% 3-class) [8]; SVD entropy + KNN at 90% overlap (F1 93% binary) [9]; recurrence complexity (AUC 0.82, 3-class) [10]; electrode-pair coherence + MI (71% 3-class, LOSOCV) [11]; FOOOF aperiodic components (AUC 0.73, AD/FTD) [12]; PLI/AEC-c graph features (AUC 0.85, 3-class) [13]; time-frequency FC + SVM (95.4% binary) [14]; CNN spectrograms (97.3% binary AD/HC) [15]; and Riemannian stacking (AUC 0.82, LOSOCV) [16]. Most high-accuracy results rely on LOSOCV or binary tasks. This work fills the gap by jointly benchmarking classical and deep-learning pipelines under a strict held-out 70/10/20 subject split on the full 3-class problem.

Contributions:

- A 1 805-feature hand-crafted pipeline (time-domain, spectral, DWT, inter-channel coherence).
- A CWT scalogram pipeline with seven deep architectures; best: OvR YOLO26m-cls + threshold optimisation at **88.89%** 3-class subject accuracy.
- Subject-wise stratified splits and majority voting ensuring clinically valid, leakage-free evaluation.

II. Materials and Methods

A. Dataset and Exploratory Data Analysis

The ds004504 dataset [4] comprises resting-state EEG from 88 subjects: 36 AD, 23 FTD, 29 HC, recorded at 500 Hz with a 19-electrode standard 10-20 montage (Fp1, Fp2, F7, F3, Fz, F4, F8, T3, C3, Cz, C4, T4, T5, P3, Pz, P4, T6, O1, O2) during closed-eyes rest. Preprocessed derivatives include band-pass filtering (0.5–45 Hz), Artifact Subspace Reconstruction (ASR), and ICA artefact removal. Signals are read via MNE-Python and scaled to a 0.1 mV amplitude range ($\approx \pm 150\,\mu$V equivalent).

Exploratory data analysis was performed on single subjects prior to segmentation. For each subject the following representations were examined on channel F4: full-signal time series, FFT magnitude and log-power spectrum, Welch periodogram, STFT spectrogram, and CWT scalogram (cmor-1.0-1.0, 60 scales) (Fig. 1). The same representations were repeated on a single 20-second segment to illustrate the analysis unit. Band-power topography maps computed via Welch PSD across all 88 subjects (Fig. 2) reveal class-discriminative spatial patterns [8], [9]: AD shows elevated posterior δ/θ and markedly reduced α; FTD exhibits frontal θ elevation with relatively preserved α; HC displays dominant occipital α.

B. Preprocessing and Epoch Segmentation

Continuous recordings were segmented into non-overlapping 20-second epochs (10 000 samples at 500 Hz). Recording durations range from $\sim$180 s to over 600 s, yielding 9–30+ epochs per subject. Total: 3 446 epochs (AD: 1 439, FTD: 815, HC: 1 192). All epochs from one subject are assigned to exactly one partition [8], [9]. Subject-wise stratified split: Train $\approx$62 (70%), Val $\approx$9 (10%), Test $\approx$17 (20%), random state 42. A runtime assertion verifies mutual disjointness of subject ID sets. Class imbalance is addressed by inverse-frequency weights: $w_c = N_{\text{total}}/(3N_c)$.

C. Hand-Crafted Feature Pipeline

A 59-dimensional feature vector is extracted per channel per epoch.

Time-domain (27): max, min, mean, variance, std, MAD, RMS, mean absolute first-difference, energy, peak-to-peak, total variation, Shannon entropy, trapezoidal integral, lag-1 autocorrelation, temporal centroid, local peak count, curve length, mean power, zero-crossing rate, skewness, kurtosis, local maxima/minima counts, amplitude range, Hjorth mobility, Hjorth complexity [8].

Frequency-domain (20): peak PSD and frequency, median frequency, spectral centroid, bandwidth, entropy, spread, skewness/kurtosis, 95th-percentile frequency, infra-low power, high-frequency ratio, spectral asymmetry, relative δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), β (13–30 Hz) powers [9].

DWT (12): five-level db4 decomposition (A5, D5–D1); total energy, wavelet entropy, five inter-level energy ratios, six sub-band variances [8], [9].

Concatenated across 19 channels: $59 \times 19 = 1\,121$ features. Inter-channel coherence for all $\binom{19}{2} = 171$ pairs in four bands adds $171 \times 4 = 684$ features [7], [11], [16]. **Total: 1 805 features/epoch.** StandardScaler + PCA (95% variance) applied train-only. Classifiers: SVM (RBF, $C=10$, balanced), Random Forest (300 trees), XGBoost (300 est., depth 6, lr 0.1). Subject predictions via majority vote.

D. CWT Scalogram and Deep Learning Pipelines

CWT scalogram. Each epoch is transformed with cmor-1.0-1.0 wavelet (60 linear scales 11–500) producing a $(19, 60, 1000)$ tensor ($10\times$ time decimation, log compression $10\log_{10}(|\cdot| + 1)$). Seven architectures: (1) Custom CNN (AlexNet-inspired, GroupNorm, SiLU); (2) EfficientNet-B0 ($3 \rightarrow 19$ ch stem patch); (3) ResNet-50 variant (BottleneckEEG, SiLU/GroupNorm); (4) DenseNet (4 blocks, growth rate 24); (5) ViT (DeiT-Tiny, patch 6×50); (6) YOLO26m-cls (first conv patched, averaged-RGB init scaled $\times 3/19$, softmax override); (7) OvR YOLO26m-cls (three binary classifiers, one per class).

OvR Baseline. Three binary YOLO26m-cls models: *model_ad*, *model_ftd*, *model_hc*. Decision: $\hat{y} = \text{argmax}[P_{\text{AD}}(1),\, P_{\text{FTD}}(1),\, P_{\text{HC}}(1)]$.

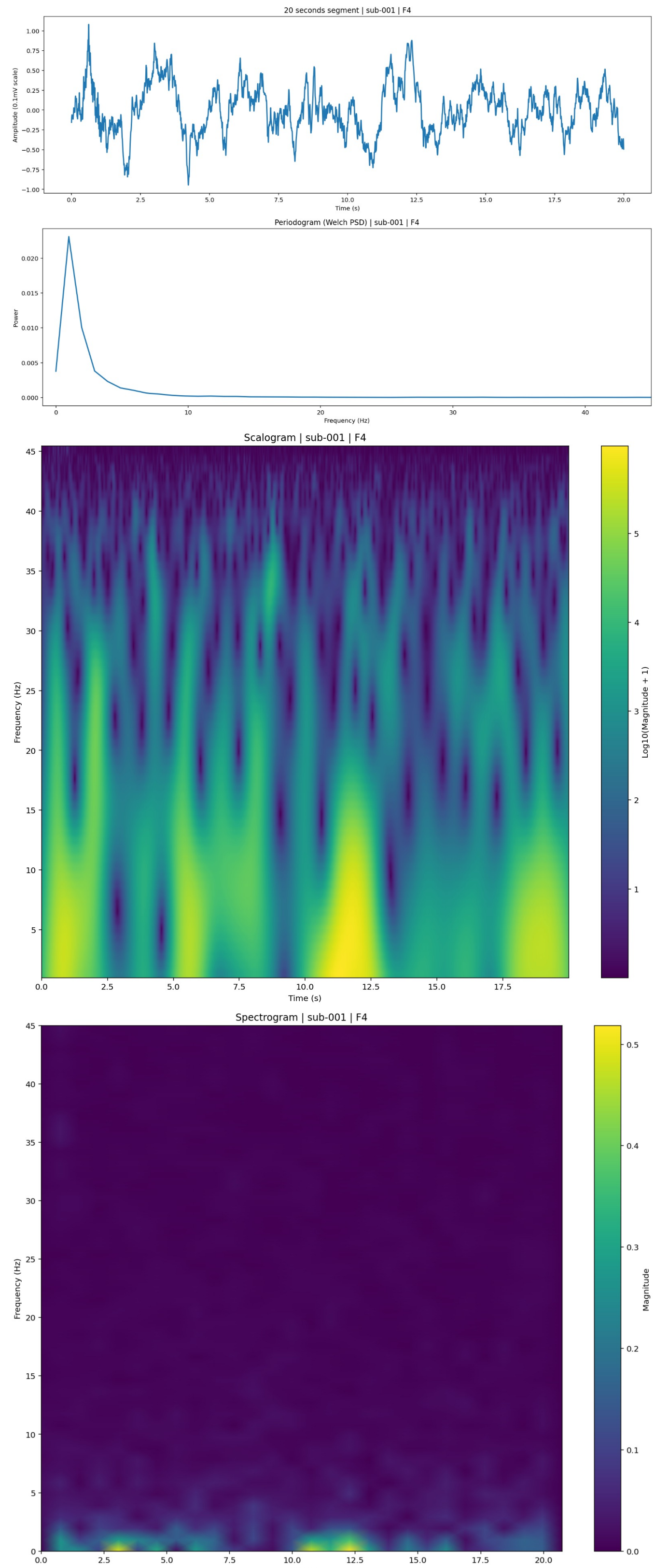

Fig. 1. EDA of sub-001 (AD), channel F4 — 20-second epoch. (A) raw time series; (B) Welch PSD; (C) CWT scalogram (cmor-1.0-1.0, log-compressed); (D) STFT spectrogram (log-power).

OvR + Threshold. Same models; per-class thresholds $(\theta_{\text{AD}}, \theta_{\text{FTD}}, \theta_{\text{HC}})$ optimised on validation set by grid search over $[0.05, 0.95]$ (step 0.05; $19^3 = 6\,859$ combinations) maximising weighted F1. Decision: $\hat{y} = \operatorname{argmax}[P_{\text{AD}} - \theta_{\text{AD}}, P_{\text{FTD}} - \theta_{\text{FTD}}, P_{\text{HC}} - \theta_{\text{HC}}]$. Typical: $\theta_{\text{AD}} \approx 0.20$–$0.40$, $\theta_{\text{FTD}} \approx 0.05$, $\theta_{\text{HC}} \approx 0.15$–$0.35$.

All scalogram models use AdamW, cosine annealing LR, weighted cross-entropy with label smoothing, gradient clipping (max_norm=2.0 for YOLO), 150–200 epochs, best weights by validation loss.

EEG-native models. Raw epochs downsampled to 128 Hz (19×2560): EEGNet [17] ($F_1 = 8$, $D = 2$, temporal kernel 250 samples); ShallowConvNet [18] (40 filters, square/log activation); SpectrogramCNN (log-power spectrograms 19×65×78, 3-layer 2D CNN).

III. Results

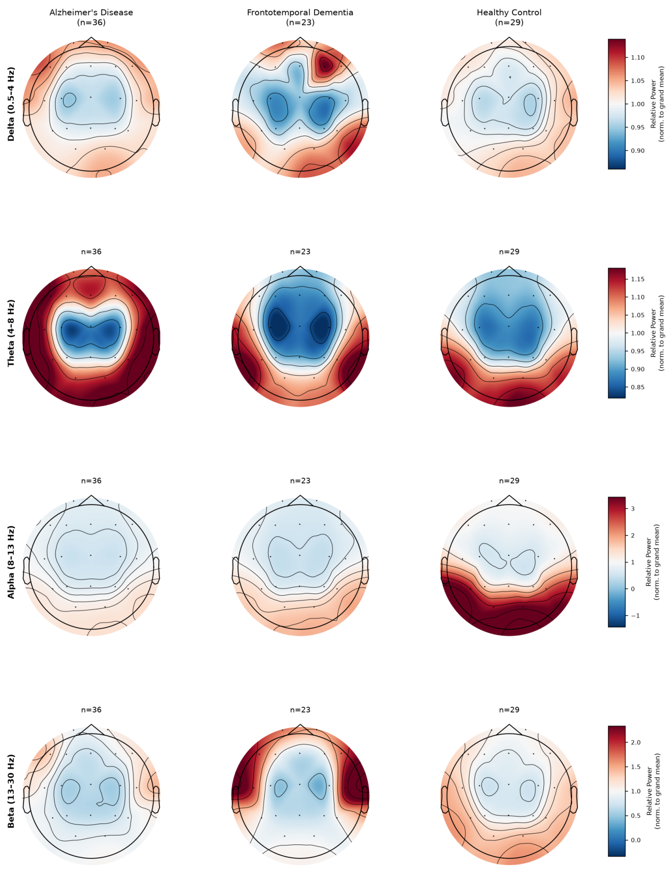

Fig. 2. Grand-mean normalised band-power topography maps (Welch PSD) across AD, FTD, and HC for four frequency bands. Colour scale relative to grand mean across all 88 subjects (1.0 = average). AD: elevated posterior δ/θ, strongly reduced α. FTD: frontal θ elevation, relatively preserved α. HC: dominant occipital α. Diverging red–blue, readable in grayscale.

Table I contextualises our results against all published studies on ds004504. Tables II and III report our test-set performance. Fig. 3 shows the subject-level confusion matrix and ROC curves for the best model. Fig. 4 compares all 15 models.

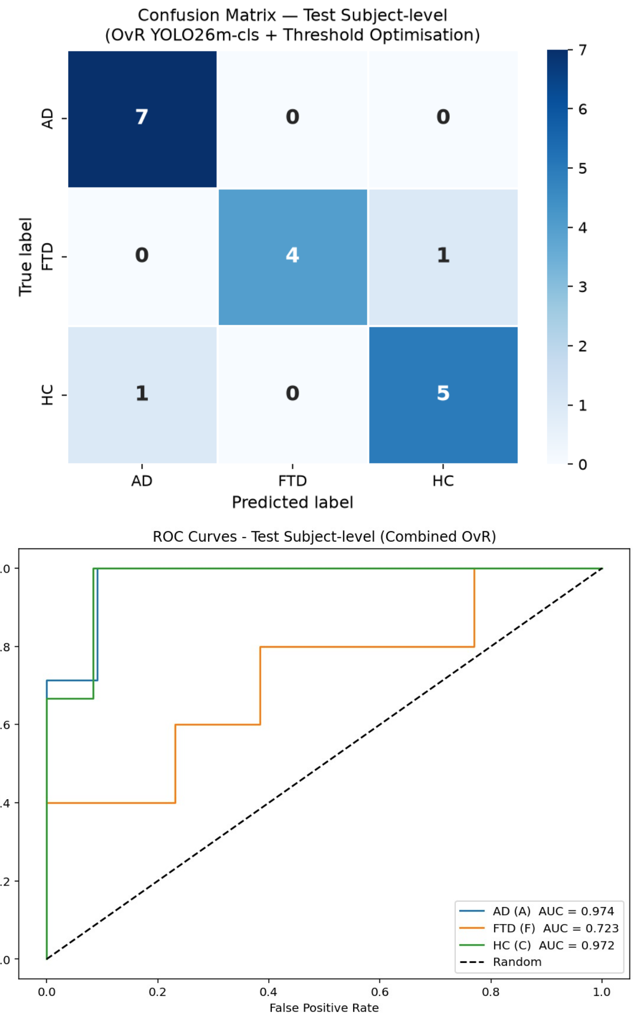

Fig. 3. Best model (OvR YOLO26m-cls + Threshold, 17 test subjects). *Top:* subject-level confusion matrix — AD: 7/7; FTD: 4/5 (1 to HC); HC: 5/6 (1 to AD). *Bottom:* subject-level ROC — AD AUC = 0.974, FTD AUC = 0.723, HC AUC = 0.972. FTD is consistently the hardest class.

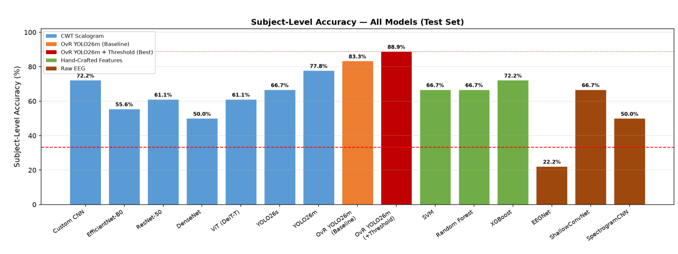

Fig. 4. Subject-level accuracy for all 15 models (held-out test set). Blue = CWT scalogram; orange = OvR baseline; red = OvR + threshold (best); green = hand-crafted; brown = raw EEG / spectrogram. Dashed line: chance (33.3%).

TABLE I
PUBLISHED RESULTS ON DS004504 EEG DATASET (EXTERNAL BENCHMARK)

Study	Method / Features	Protocol	Task	Best Result
Miltiadous et al. [4] (2023)	Relative Band Power + SVM/KNN	Full signal	Binary	∼70–75% Acc
Miltiadous et al. [5] (2023)	STFT + DICE-Net (Conv-Transformer)	–	Binary / 3-class	96.3% / ∼83% Acc
Si et al. [6] (2023)	PLV multilayer graph + SVM	2 s epochs	AD/FTD binary	89.6% Acc, AUC 0.88
Ajra et al. [7] (2023)	AEC + Shallow CNN	–	Binary	94.54% Acc
Rostamikia et al. [8] (2024)	Lyapunov Exponents + SVM	10 s epochs	3-class	90.2% Acc
Lal et al. [9] (2024)	SVD Entropy + KNN	2 s / 90% overlap	Binary	F1 93%, Acc 91%
Zheng et al. [10] (2024)	MTRRP + SVM	Full signal	3-class	AUC 0.82
Ma et al. [11] (2024)	Coherence + MI + SVM	LOSOCV	3-class	∼71% Acc
Wang et al. [12] (2024)	FOOOF aperiodic + SVM	MCCV + SMOTE	AD/FTD binary	AUC 0.73
Wu et al. [13] (2024)	PLI / AEC-c graph + SVM	4 s epochs	3-class	AUC 0.85
Zheng et al. [14] (2025)	Time-freq FC + SVM	LOSOCV, 4 s	Binary	95.4% Acc
Stefanou et al. [15] (2025)	STFT CNN (VGG/ResNet)	30 s epochs	Binary AD/HC	97.3% Acc
Mlinaric et al. [16] (2026)	Riemannian FC stacking	LOSOCV	Binary	AUC 0.82
This work	CWT OvR YOLO26m + Threshold	**70/10/20 held-out**	**3-class**	**88.89% Subj. Acc**

LOSOCV = Leave-One-Subject-Out CV. MCCV = Monte Carlo CV. Most >90% results use LOSOCV or binary tasks; not directly comparable to our held-out 3-class s

TABLE II
HAND-CRAFTED FEATURE PIPELINE — TEST SET RESULTS

Model	Seg. Acc.	Subj. Acc.	Subj. F1	AUC
SVM (RBF, $C = 10$)	63.3%	66.7%	52.3%	85.3%
Random Forest (300 trees)	60.2%	66.7%	51.6%	76.1%
XGBoost (300, depth 6)	60.6%	**72.2%**	**64.8%**	73.6%

TABLE III
DEEP LEARNING RESULTS — TEST SET (SUBJECT-LEVEL / MAJORITY VOTE)

Input	Model	Subj. Acc.	Subj. F1
CWT Scalogram $(19, 60, 1000)$	Custom CNN	72.22%	67.04%
	EfficientNet-B0	55.56%	55.31%
	ResNet-50 variant	61.11%	59.03%
	DenseNet	50.00%	50.00%
	ViT (DeiT-Tiny)	61.11%	58.06%
	YOLO26s-cls	66.67%	62.38%
	YOLO26m-cls	77.78%	77.08%
CWT + OvR (3×YOLO26m)	OvR Baseline	83.33%	82.64%
	OvR + Threshold	**88.89%**	**88.77%**
Raw EEG 128 Hz $(1, 19, 2560)$	EEGNet [17]	22.2%	18.9%
	ShallowConvNet [18]	66.7%	52.3%
	SpectrogramCNN	50.0%	38.9%

Classical methods. XGBoost achieved the highest subject accuracy (72.2%) and F1 (64.8%), benefiting from the 684 coherence features [9]. SVM attained the highest AUC (85.3%) [6], [8].

CWT scalogram. OvR Baseline improved over direct YOLO26m-cls from 77.78% to 83.33% (+5.55%). OvR + Threshold further improved to **88.89%** (+5.56%), demonstrating that per-class threshold optimisation on the validation set is highly effective for subject-level majority-vote aggregation. Standard pretrained backbones underperformed due to domain shift from RGB images.

EEG-native models. ShallowConvNet (66.7%) outperformed EEGNet (22.2%) [18]. EEGNet's failure is attributed to its design for short (<4 s) binary windows; its temporal kernel covers only ∼2 s of the 20 s epoch.

FTD recall is the weakest across all models (binary FTD recall 34.78%–40.22%), consistent with all literature on this dataset [4], [9].

IV. DISCUSSION AND CONCLUSION

This study demonstrates that 3-class AD/FTD/HC separation from resting-state EEG is feasible under strict held-out evaluation [8], [13], [14].

The OvR + Threshold approach (88.89%) is the best result. Decomposing the 3-class problem into three binary tasks with per-class threshold tuning addresses both the hard AD/FTD boundary [6], [11], [12], [16] and FTD class imbalance (23 subjects). The OvR Baseline (83.33%) already matches DICE-Net's reported 3-class result (∼83%) [5] despite a stricter fixed split.

XGBoost on 1 805 hand-crafted features (72.2%) is competitive with most scalogram models and fully interpretable. The 684 inter-channel coherence features are crucial [7], [11], [16].

Limitations. The 17-subject test set limits statistical precision. FTD recall is the weakest throughout [4], [9]. EEGNet is poorly matched to 20 s multi-class epochs. Our fixed 70/10/20 split is more conservative than LOSOCV; direct numerical comparison with LOSOCV papers should account for this.

Future work. Augmentation (scalogram mixup, time-warping), Grad-CAM explainability [5], aperiodic EEG components [12], and recurrence complexity features [10] are natural extensions.

Conclusion. We benchmarked 15 pipelines for 3-class EEG dementia classification. OvR YOLO26m-cls + threshold optimisation achieved **88.89% subject accuracy (F1 = 88.77%)** under a conservative held-out split. XGBoost on hand-crafted features provided a strong baseline (72.2%). Subject-wise stratification, majority voting, and held-out evaluation are essential for clinically meaningful benchmarking.

V. PRESENTATION LINK

The recorded presentation is available at: [INSERTYOUTUBEORDRIVELINKHERE]

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